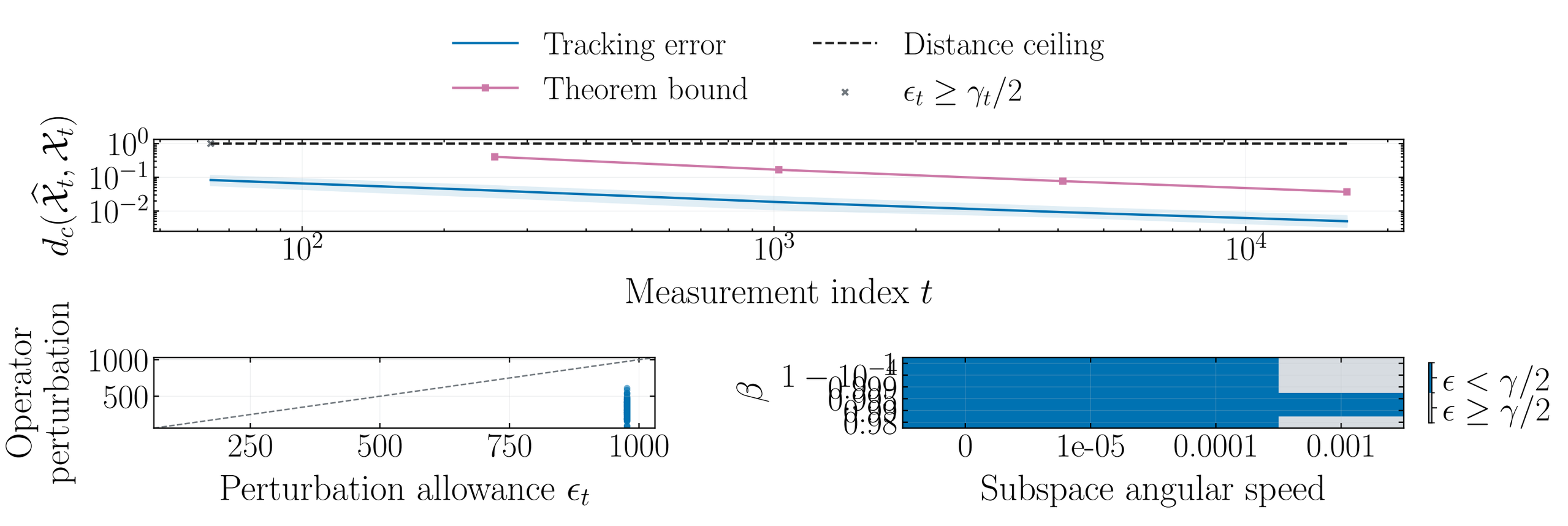

Tracking error
Theorem bound
Distance ceiling
$\epsilon_t \geq \gamma_t/2$
$d_c(\hat{\mathcal{X}}_t, \mathcal{X}_t)$
10^0
10^{-1}
10^{-2}
10^2
10^3
10^4
Measurement index t
Operator perturbation
1000
500
250
500
750
1000
Perturbation allowance ϵ_t
β
$1 - 10^{-1}$
0.999
0.998
0.997
0.996
0.995
0.994
0.993
0.992
0.991
0.99
0.989
0.988
0.987
0.986
0.985
0.984
0.983
0.982
0.981
0.98
0
1e-05
0.0001
0.001
Subspace angular speed
$\epsilon < \gamma/2$
$\epsilon \geq \gamma/2$